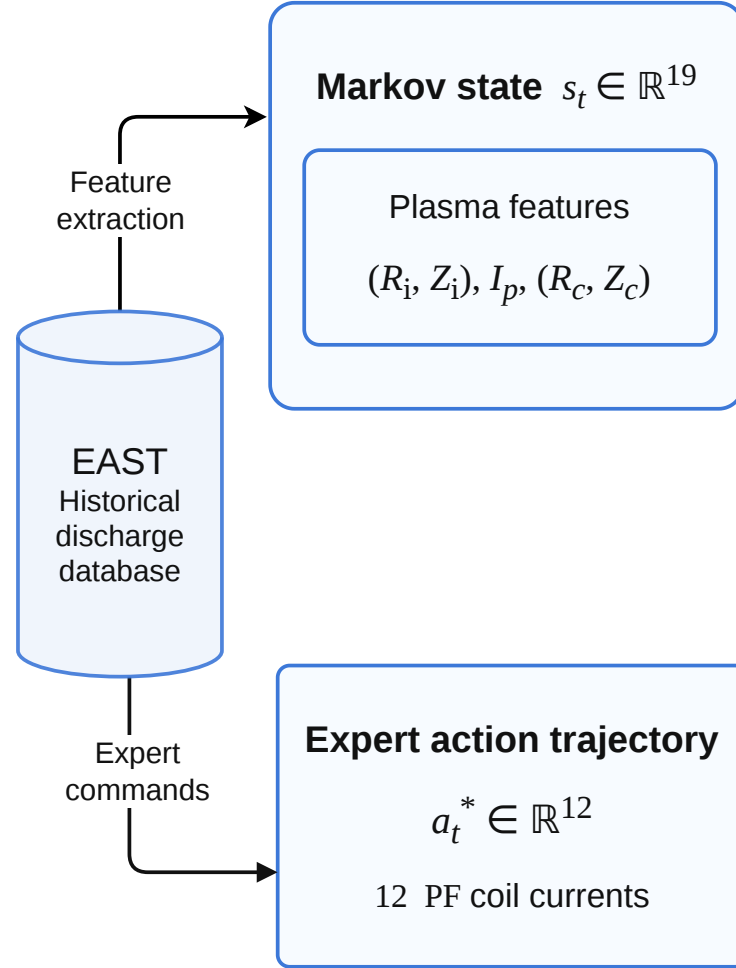
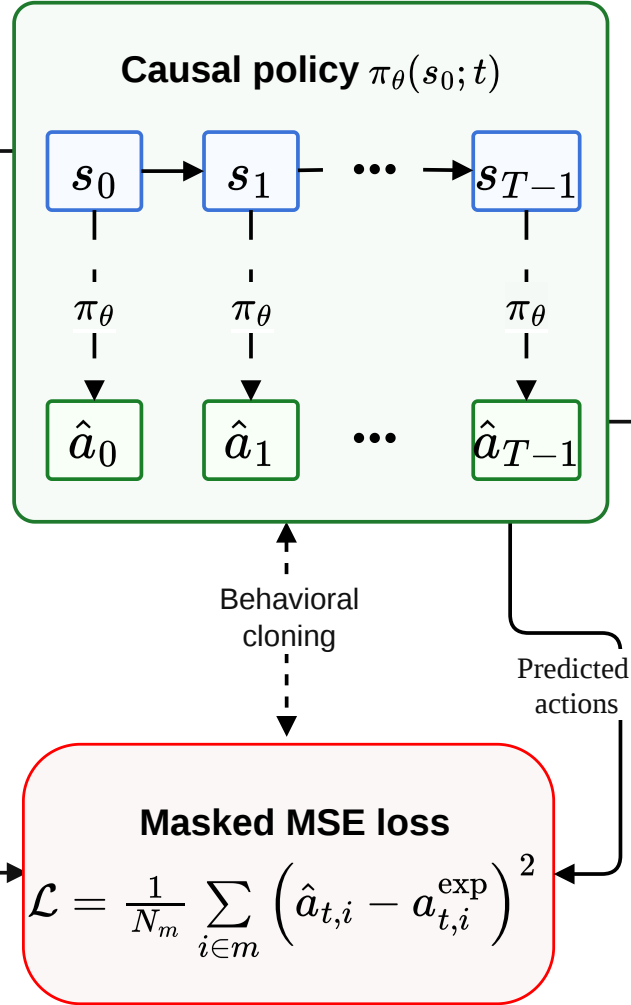


a Expert Demonstrations



b Behavioral Cloning



c Operational Constraints

1) Current limit

$$|I_c| \leq 14.5 \text{ kA}$$

2) Rate of current change

$$\left| \frac{dI_c}{dt} \right| \leq \tau_{p,c}$$

Safe control action

$$a_t^{(\text{safe})}$$

d Physics consistency Validation

Fixed Plasma Profiles from EFIT

$$p'(\psi), FF'(\psi)$$

FreeGSNKE

$$\Delta^* \psi = -\mu_0 R^2 p'(\psi) - FF'(\psi)$$

EFIT (R_8, Z_8) FreeGSNKE $I_{\text{Pred}}(R_8, Z_8)$

